2+2

Probability :-

Date: _____

Definition :-

A phenomena is called random if the exact outcome is uncertain. The mathematical study of randomness is called the "Theory of Probability".

Example :-

If a coin is tossed in the air, it is certain that the coin will come down, but it is not certain whether head will occur.
Thus probability is that branch of science which deals with chances.

Extra :-

- Probability tells us how likely an event to happen.
- Always between 0 (impossible) and 1 (certain).
- More the chance → probability closer to 1.

"Probability is define as the measure of uncertainity. It is the branch of science which deals with chances."

SP PAPER PRODUCT

Scanned with CamScanner

Axioms of Probability :-

Date:_____

Let S be the sample space and E any event in S.

Axiom 1: If an event is certain not to occur, then its probability is

$$P(\emptyset) = 0$$

Axiom 2: If an event is certain to occur, then its probability is

$$P(S) = 1$$

Axiom 3: From axiom 1 and 2, we can say that for every event E:

$$0 \leq P(E) \leq 1$$

Axiom 4: For two events E_1 and E_2.

- If both are mutually exclusive, then

$$P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

- If they are not mutually exclusive, then

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

SP PAPER PRODUCT

Scanned with CamScanner

Law of Addition of Probability For Any Two Events :-

Date: _____

Statement: If E_1 and E_2 are two non-mutually exclusive events defined in sample space S, then the probability that E_1 occur or E_2 occur or both of them occur is given by:

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2) \longrightarrow (1)$$

Proof:-

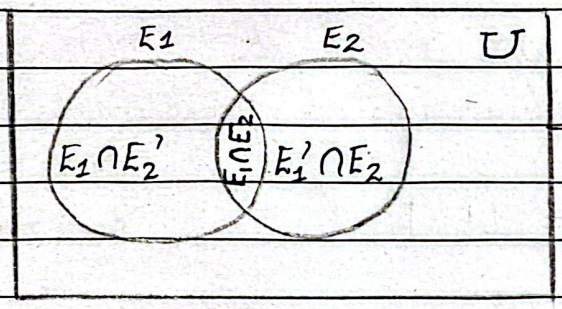


Fig 1.

From fig 1.

$$E_1 \cup E_2 = E_1 \cup (E_1' \cap E_2)$$

$$P(E_1 \cup E_2) = P[E_1 \cup (E_1' \cap E_2)]$$

$$= P(E_1) + P(E_1' \cap E_2)$$

$$= P(E_1) + P(E_1' \cap E_2) + P(E_1 \cap E_2) - P(E_1 \cap E_2)$$

$$= P(E_1) + P[(E_1' \cap E_2) \cup (E_1 \cap E_2)] - P(E_1 \cap E_2) \quad \rightarrow E_2$$

$$= P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

Proved

PAPER PRODUCT

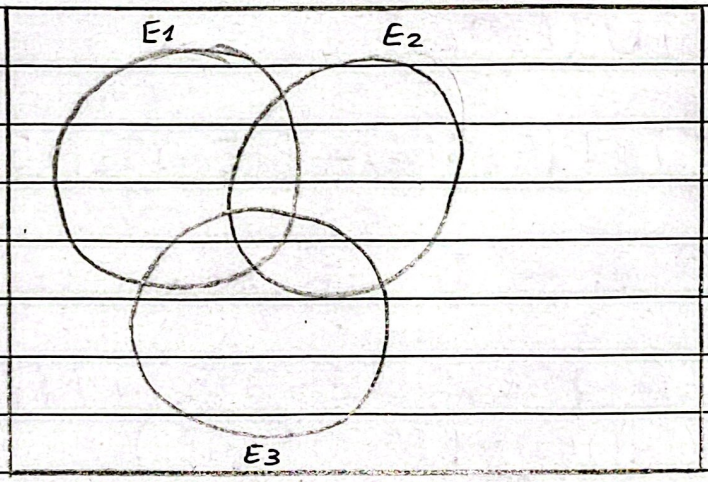
Scanned with CamScanner

Law of Addition of Probability For Any Three Events E_1, E_2 & E_3 :- Date:_____

Statement: If E_1, E_2 and E_3 are three non-mutually exclusive events defined in the sample space S, then the probability that 'E_1 or E_2 or E_3' occures or the probability that atleast one of them occurs is given by:

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_1 \cap E_3) - P(E_2 \cap E_3) + P(E_1 \cap E_2 \cap E_3) \longrightarrow (1)$$

Proof:



Let $E = E_1 \cup E_2$

$$\Rightarrow P(E) = P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2) \longrightarrow (2)$$

Let E and E_3 be two events then according to law of two non-mutually exclusive events we have.

$$\Rightarrow P(E \cup E_3) = P(E) + P(E_3) - P(E \cap E_3) \longrightarrow (3)$$

PAPER PRODUCT

Scanned with CamScanner

Date:_____

By substituting the value of E in above eqⁿ, we have

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) - P(E_1 \cap E_2) + P(E_3) - P(E \cap E_3) \longrightarrow (4)$$

Now consider,

$$E \cap E_3 = (E_1 \cup E_2) \cap E_3 \quad \text{, By distributive law}$$

$$= (E_1 \cap E_3) \cup (E_2 \cap E_3)$$

$$P(E \cap E_3) = P(E_1 \cap E_3) + P(E_2 \cap E_3) - P(E_1 \cap E_2 \cap E_3) \longrightarrow (5)$$

Substituting eqⁿ (5) in eqⁿ (4), we get:

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) - P(E_1 \cap E_2) + P(E_3) - P(E_1 \cap E_3) - P(E_2 \cap E_3) + P(E_1 \cap E_2 \cap E_3).$$

Rearrange the terms to obtain the final formula:

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_1 \cap E_3) - P(E_2 \cap E_3) + P(E_1 \cap E_2 \cap E_3).$$

Proved

This is the required result.

PAPER PRODUCT

Scanned with CamScanner

Independent Events :-

Date:_____

Two non-mutually exclusive events are said to be independent if the occurence of one does not affect on the occurence of the other.

Mathematically :-

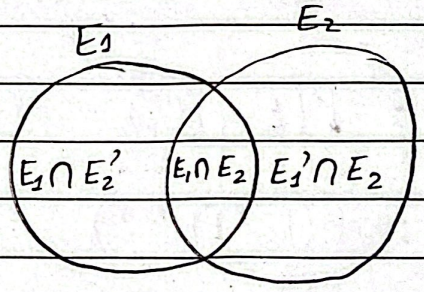
$$P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$$
$$P(E_1 \cap E_2 \cap E_3) = P(E_1) \cdot P(E_2) \cdot P(E_3)$$

Theorem 1 : If E_1 and E_2 are independent then show that E_1 and E_2' are also independent.

Given : $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2) \longrightarrow (1)$

To Prove : $P(E_1 \cap E_2') = P(E_1) \cdot P(E_2') \longrightarrow (2)$

From venn diagram, we have



Take L.H.S of eqn (2).

$$P(E_1 \cap E_2') = P(E_1) - P(E_1 \cap E_2) \rightarrow (3)$$

Substituting eqn (1) in eqn (3).

$$P(E_1 \cap E_2') = P(E_1) - P(E_1) \cdot P(E_2)$$
$$= P(E_1)[1 - P(E_2)]$$
$$= P(E_1) \cdot P(E_2')$$

Proved

PAPER PRODUCT

Scanned with CamScanner

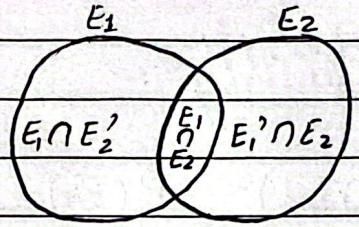
Theorem 2:

Date:_____

Given: $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$ ⟶ (1)

To Prove: $P(E_1' \cap E_2) = P(E_1') \cdot P(E_2)$ ⟶ (2)

From venn diagram, we have.



$P(E_1' \cap E_2) = P(E_2) - P(E_1 \cap E_2)$ ⟶ (3)

Substituting eq^n (1) in eq^n (3), we get

$$P(E_1' \cap E_2) = P(E_2) - P(E_1) \cdot P(E_2)$$
$$= P(E_2)[1 - P(E_1)]$$
$$= P(E_1')(E_2)$$

Proved

Theorem 5: If 2 events E_1 and E_2 are independent then E_1' and E_2' are also independent.

Given: $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$ ⟶ (1)

Proof: $P(E_1' \cap E_2') = P(E_1') \cdot P(E_2')$ ⟶ (2)

By Demorgan's law, we have.

$$P(E_1' \cap E_2') = P(E_1 \cup E_2)'$$
$$= 1 - P(E_1 \cup E_2)$$
$$= 1 - [P(E_1) + P(E_2) - P(E_1 \cap E_2)]$$
$$= 1 - P(E_1) - P(E_2) + P(E_1 \cap E_2)$$ ⟶ (3)

PAPER PRODUCT

Scanned with CamScanner

Date:_____

Substituting eq^{n} ② in eq^{n} ③, we get:

$$P(E_1' \cap E_2') = 1 - P(E_1) - P(E_2) + P(E_1)(E_2)$$
$$= [1 - P(E_1)] - [1 - P(E_1)]\, P(E_2)$$
$$= [1 - P(E_1)] \cdot [1 - P(E_2)]$$
$$= P(E_1') \cdot P(E_2')$$

Proved

Conditional Probability :-

Definition :-

The probability of an event depending on the outcome of another event is called conditional probability. When some additional information about the experiment is known, the sample space reduces. The probabilities with respect to this reduced sample space are called conditional probabilities.

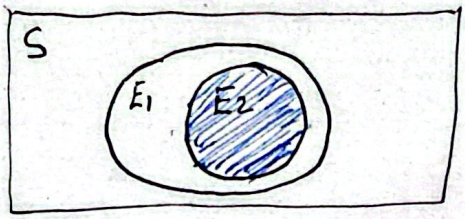
Mathematical Definition :-

Let S be the sample space, and let E_1 and E_2 be two events defined in S. The probability that E_2 occurs given that E_1 has already occured is denoted by:

$$P(E_2|E_1) = \frac{P(E_1 \cap E_2)}{P(E_1)}, \quad P(E_1 > 0)$$

SP PAPER PRODUCT

Scanned with CamScanner



Date:

or equivalently,

$$P(E_1 \cap E_2) = P(E_1) \cdot P(E_2 | E_1)$$

Similarly,

$$P(E_1 | E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)}, \quad P(E_2) > 0$$

and

$$P(E_1 \cap E_2) = P(E_2) \cdot P(E_1 | E_2)$$

Special Case (Independent Events):

If E_1 and E_2 are independent, then.

$$P(E_2 | E_1) = P(E_2), \quad P(E_1 | E_2) = P(E_1)$$

So,

$$P(E_1 \cap E_2) = P(E_1) \cdot P(E_2).$$

SP PAPER PRODUCT

Scanned with CamScanner

Baye's Theorem

Date: _____

Statement :-

If the events $E_1, E_2, \ldots, E_n$ constitute a partition of the sample space S, where $P(E_i) \neq 0$ for $i = 1, 2, \ldots, n$, then for any event E defined in the sample space S such that $P(E) > 0$, the probability of E_k occuring given that event E has occured is given by:

$$P(E_k|E) = \frac{P(E_k) \cdot P(E|E_k)}{P(E_1) \cdot P(E|E_1) + P(E_2) \cdot P(E|E_2) + \cdots + P(E_n) \cdot P(E|E_n)}$$

OR written as a summation:

$$P(E_k|E) = \frac{P(E_k) \cdot P(E|E_k)}{\sum_{i=1}^{n} P(E_i) \cdot P(E|E_i)}$$

Proof :-

Suppose S be the sample space and let events $E_1, E_2, \ldots, E_n$ be the partitions (mutually exclusive events) of the sample space S. Let E be any other event (subset of S) such that $P(E) > 0$. Then:

$$E = E \cap S$$

$$E = E \cap (E_1 \cup E_2 \cup E_3 \cup \ldots \cup E_n)$$

PAPER PRODUCT

Scanned with CamScanner

Date:

By Distributive law, we have:

$$E = (E \cap E_1) \cup (E \cap E_2) \cup (E \cap E_3) \cup \dots \cup (E \cap E_n)$$

By addition law of mutually exclusive events:

$$P(E) = P(E \cap E_1) + P(E \cap E_2) + \dots + P(E \cap E_n) \longrightarrow (1)$$

By the definition of Conditional Probability:

$$P(E \cap E_1) = P(E_1) \cdot P(E|E_1)$$
$$P(E \cap E_2) = P(E_2) \cdot P(E|E_2) \quad \dots \text{ and so on.}$$

Substituting these values, we get the total probability of E.

$$P(E) = P(E_1) \cdot P(E|E_1) + P(E_2) \cdot P(E|E_2) + \dots + P(E_n) \cdot P(E|E_n) \longrightarrow (2)$$

For any k^{th} event, the conditional probability of E_k given E is defined as:

$$P(E_k|E) = \frac{P(E_k \cap E)}{P(E)}$$

Substituting the value of P(E) from eqn (2).

$$P(E_k|E) = \frac{P(E_k) \cdot P(E|E_k)}{P(E_1) \cdot P(E|E_1) + P(E_2) \cdot P(E|E_2) \dots + P(E_n) \cdot P(E|E_n)}$$

Proved

PAPER PRODUCT

Scanned with CamScanner